

Prototype and Implementation Plan

Project: Sahayak - Welfare Scheme Discovery Assistant
Live demo: https://parzival1821.github.io/Multilingual_Chatbot/
Repository: https://github.com/parzival1821/Multilingual_Chatbot

Prototype Status

The current prototype is a deployed static web app. It demonstrates the end-to-end user journey required for the NSS Challenge 1.1 MVP:

1. Select English or Hindi.
2. Fill an eligibility profile.
3. View personalized welfare scheme recommendations.
4. Ask grounded follow-up questions.
5. Generate, copy, and download a document checklist.

MVP Feature Map

Feature	Status	Notes
Live prototype	Complete	GitHub Pages deployment
8-scheme catalogue	Complete	Central schemes with official source links
Eligibility questionnaire	Complete	Uses household and occupation signals
Personalized recommendations	Complete	Ranked cards with reasons
Hindi interface	Complete	Devanagari UI, prompts, scheme names, summaries
Follow-up assistant	Complete	Retrieval-grounded answer generation
Safe fallback	Complete	Prevents unsupported answers
Document checklist	Complete	Copy/download text checklist
Demo personas	Complete	Farmer, woman-led household, street vendor
Automated tests	Complete	Core and static smoke checks

Repository Structure

```
.
├── index.html
├── styles.css
├── package.json
├── README.md
├── src
│   ├── app.js
│   ├── core.js
│   └── data.js
├── tests
│   ├── core.test.js
│   └── static-smoke.test.js
├── submission
│   ├── 01-problem-analysis-report.md
│   ├── 02-solution-framework.md
│   ├── 03-prototype-implementation-plan.md
│   ├── 04-impact-projection.md
│   ├── 05-pilot-test-report.md
│   ├── 06-final-presentation-outline.md
│   ├── 07-submission-portal-details.md
│   └── 08-visual-diagrams.md
```

Key Files

File	Responsibility
index.html	Dashboard structure
styles.css	Responsive visual design
src/data.js	Scheme knowledge base and language labels
src/core.js	Recommendation, retrieval, profile inference, checklist formatting
src/app.js	UI event handling and rendering
tests/core.test.js	Behavioral tests for recommendations, assistant, checklist
tests/static-smoke.test.js	Static app and deployment checks

Implementation Flow

Diagram: Rendered in the Visual Diagrams Appendix PDF.

Local Run Instructions

`npm run dev`

Then open:

`http://localhost:4173`

Test Instructions

Run core tests:

`npm test`

Run all checks:

`npm run check`

Expected result:

All core tests passed.

Static smoke checks passed.

Deployment

The project is deployed using GitHub Pages from the main branch.

Deployment workflow:

1. Push changes to main.
2. GitHub Actions builds the static site.
3. Pages serves the app at the live demo URL.

Demo Script

1. Open the live app.
2. Use English mode first.
3. Click the Farmer preset.
4. Show PM-KISAN as a strong match.
5. Click View documents.
6. Copy/download the checklist.
7. Ask: What documents do I need for Ayushman Bharat?
8. Switch to Hindi.
9. Ask: मुझे घर के लिए कौन सी योजना मिल सकती है?
10. Show Hindi recommendations, source links, and fallback behavior.

Prototype Validation

Automated checks currently cover:

- PM-KISAN ranking for farmer profile.
- PM-JAY retrieval for Ayushman document questions.
- Hindi Devanagari retrieval.
- Widow follow-up routing to NSAP.
- PM SVANidhi ranking for street vendor profile.
- Checklist generation and formatting.
- Safe fallback for unsupported queries.
- Static asset references.
- GitHub Pages workflow shape.

Implementation Roadmap

Immediate Submission Version

- Keep the current 8-scheme web MVP.
- Use README and visual appendix diagrams for architecture and user flow.
- Submit GitHub repo and live demo link.
- Use the internal validation report, then add real-user observations if time permits.

Next Version

- Add one more regional language beyond Hindi.
- Add WhatsApp/SMS interface through Twilio or WhatsApp Business API.
- Add admin tool for updating scheme data.
- Expand scheme coverage using official scheme catalogues.
- Add analytics for NGO/NSS outreach camps.

Conclusion

The current prototype is ready as a functional MVP. It demonstrates the core end-to-end solution loop and is simple enough to deploy, test, and explain during evaluation.